Credit-Market Sentiment and the Business Cycle: A Replication of López-Salido, Stein & Zakrajšek (2017)

Fernando Raffo Pedraza Bangjie Xu

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Abstract

We replicate the core results of López-Salido, Stein, and Zakrajšek (2017), which shows that elevated credit-market sentiment—"frothy" credit conditions—forecasts a slowdown in real activity roughly two years ahead. This document describes the project at a high level, explains the data sources, reviews where the replication succeeded and where it was difficult, and collects every figure and table our code produces across three sets: the original replication sample, a sample extended to the present, and an Aaa-spread extension.

1 Overview of the Project

The paper's central claim is that the mood of credit markets carries information about the future path of the business cycle. When the Baa–Treasury spread is unusually narrow and a large share of new issuance is low-quality, investors are pricing credit risk aggressively; those conditions mean-revert, and spreads widen back out as activity turns down. Our code reproduces the paper's key exhibits—the long-run credit spread, the predictive regressions of output growth on credit variables, and the sentiment measure built from the Greenwood–Hanson high-yield issuance share—and reruns each on data extended to the present and on an Aaa-based variant.

2 Data Sources

Every series is pulled programmatically so the results are reproducible:

- **Federal Reserve (FRED).** Moody's seasoned Baa and Aaa corporate bond yields, the 10-year Treasury yield, real GDP, and related macroeconomic series, forming the credit spreads and output outcomes. Some of these series are only available in a newer vintage than the one used in the original paper, and several require stitching together multiple FRED series to obtain a continuous long history. The paper's appendix does not always specify exactly which underlying series were used, so where the mapping was ambiguous we made a documented judgement call on which series to combine; this is a further source of small differences from the published numbers.
- **Robert Shiller's website.** Long-run stock-market and interest-rate data, including the cyclically adjusted price-earnings ratio used as the equity-return predictor.
- **Greenwood & Hanson high-yield issuance share.** The share of high-yield issuance in total corporate bond issuance, underlying the credit-market sentiment measure. The historical portion (1926–2015) is pulled directly from its published source, the HBS Behavioral

Finance and Financial Stability `InvestorCreditSentiment` workbook, and spliced with a reconstruction from Mergent FISD (via WRDS) for recent years. The HBS workbook is a newer vintage than the values printed in the original paper's table, so some replicated numbers differ slightly from the published ones; we use the updated vintage deliberately, as it reflects the authors' latest revisions.

3 Summary Statistics of the Underlying Data

Before any regression, we document the four processed data files that feed the pipeline. Each table reports the distribution of the series in its display units; each figure is the corresponding featured chart. These correspond to the walkthrough in `01_summary_statistics.ipynb`.

Credit spreads (monthly FRED series).

FRED Monthly Series, 1925:01–2025:12 ($N = 1,212$)								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
Baa-Treasury spread	pp.	1.98	0.95	0.29	1.23	1.92	2.48	7.24
Aaa-Treasury spread	pp.	0.88	0.54	-0.17	0.39	0.81	1.23	2.68
Baa corporate bond yield	pct.	6.75	2.87	2.94	4.80	6.01	8.34	17.18
Aaa corporate bond yield	pct.	5.64	2.71	2.14	3.57	4.74	7.36	15.49
10-year Treasury yield	pct.	4.77	2.75	0.62	2.65	3.92	6.29	15.32
Months in NBER recession	pct. of months	16.6	37.2	0.0	0.0	0.0	0.0	100.0

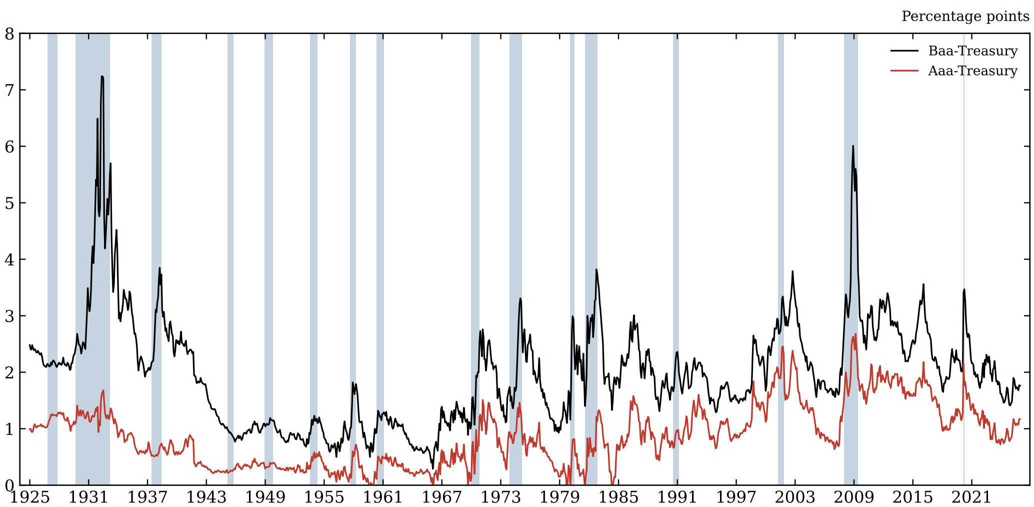

Figure 1: Baa– and Aaa–Treasury credit spreads, NBER recessions shaded.

Real GDP per capita growth (annual FRED series).

FRED Annual Series, 1929–2025 ($N = 97$)								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
Real GDP	$tn	8.68	6.72	0.88	2.88	6.68	14.23	23.85
Population	millions	223.7	69.6	121.7	159.6	220.3	285.2	341.9
Real GDP per capita	$000s	33.3	18.1	7.0	18.1	30.3	49.9	69.7
CPI inflation (Dec/Dec)	pct.	3.03	3.77	-10.84	1.49	2.71	4.00	16.66
Baa corporate bond yield	pct.	6.82	2.92	3.10	4.81	6.15	8.43	16.55
Aaa corporate bond yield	pct.	5.65	2.72	2.26	3.51	4.74	7.25	14.23
10-year Treasury yield	pct.	4.77	2.72	0.93	2.51	4.03	6.30	13.72
3-month Treasury yield	pct.	3.46	3.20	0.01	0.39	3.06	5.28	16.35
Baa-Treasury spread	pp.	2.05	1.06	0.40	1.26	1.92	2.56	6.49
Aaa-Treasury spread	pp.	0.88	0.55	-0.11	0.39	0.81	1.25	2.63

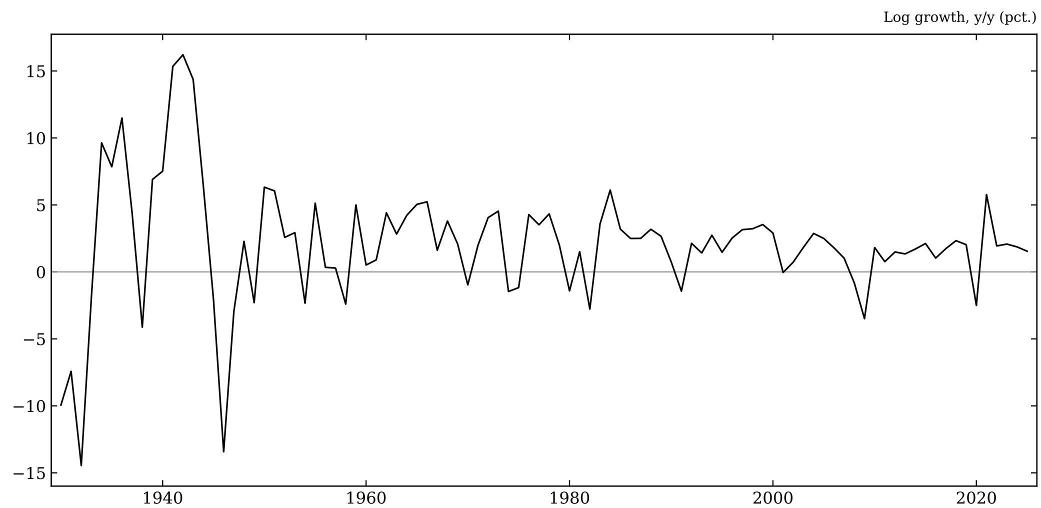

Figure 2: Growth in real GDP per capita, the dependent variable throughout Tables I and II.

High-yield issuance share (Greenwood–Hanson, spliced).

Greenwood-Hanson High-Yield Share, 1929–2025 ($N = 97$)								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
High-yield issuance share	pct.	19.5	15.1	0.0	6.2	15.0	29.4	63.9
Log high-yield share	log	-2.06	1.13	-6.02	-2.69	-1.89	-1.22	-0.45

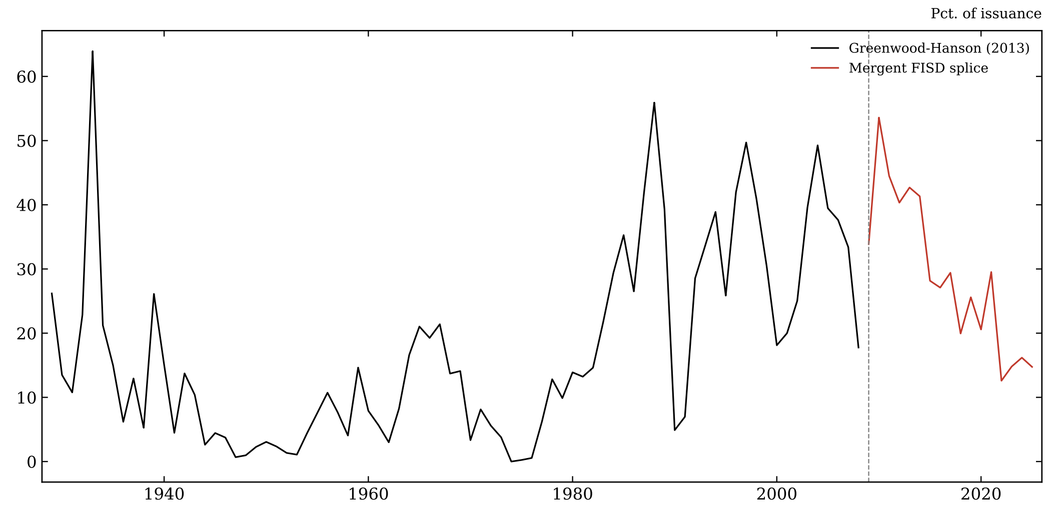

Figure 3: The high-yield share of nonfinancial corporate bond issuance, colored by source to make the 2008/2009 splice visible.

Cyclically adjusted price-earnings ratio (Shiller CAPE).

Shiller Annual Stock-Market Series, 1929–2025 ($N = 97$)								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Ma
S&P Composite price index	index level	758.66	1,318.14	6.82	26.04	106.50	1,110.38	6,853.0
S&P Composite dividend	$ per share	13.83	19.10	0.44	1.47	4.67	16.27	79.5
Cyclically adj. P/E (CAPE)	ratio	19.23	8.48	7.83	11.75	17.56	24.86	44.2
S&P total return (log)	pct.	9.54	18.06	-51.58	0.23	14.26	21.11	42.2

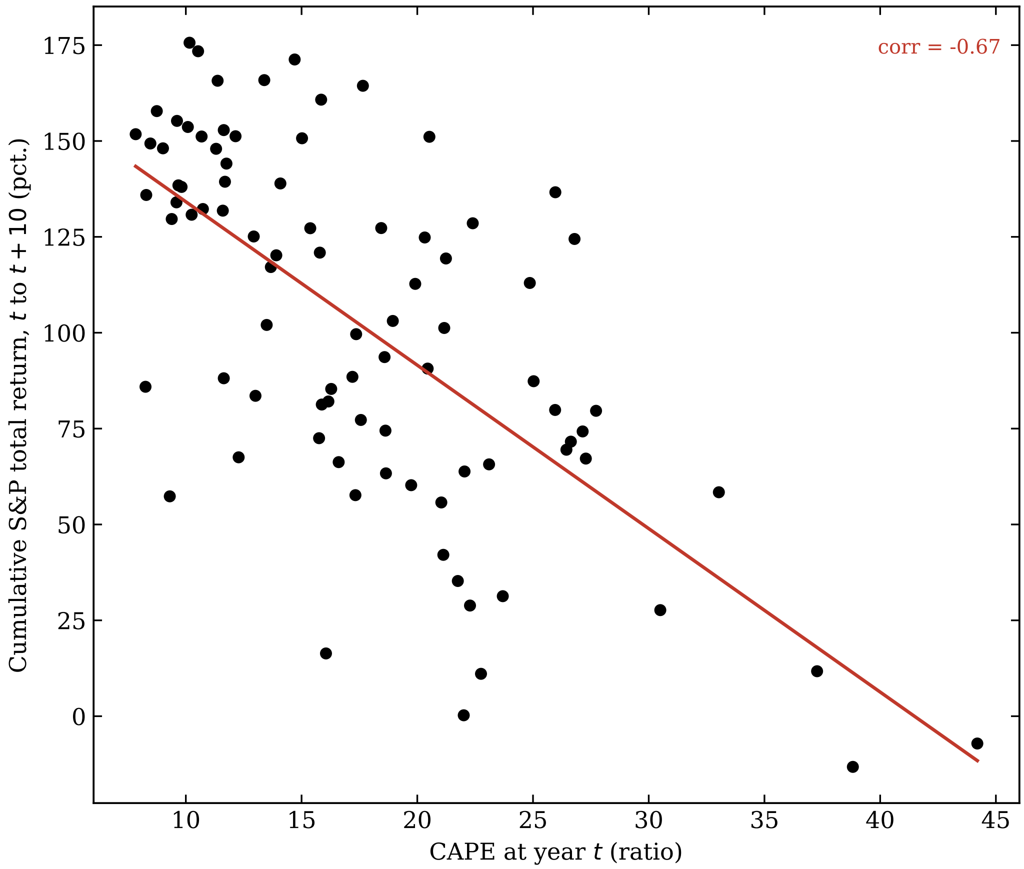

Figure 4: CAPE at year t against the cumulative S&P total return realized over the following decade.

4 Replication Sample (1929–2015)

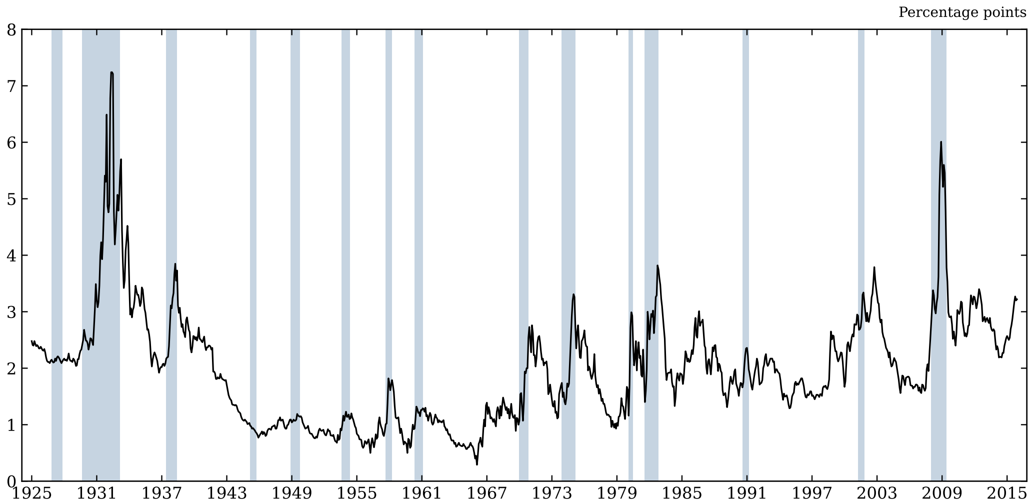

Figure 5: Figure I (replication): the Baa–Treasury credit spread, NBER recessions shaded.

Table I (replication).

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.958***	—	-2.124***
	(0.585)		(0.578)
r_t^{SP}	—	0.075**	0.022
		(0.033)	(0.035)
Δy_{t-1}	0.476***	0.478***	0.464***
	(0.104)	(0.136)	(0.114)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.248
			(0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.814**
			(0.360)
π_{t-1}	—	—	0.095
			(0.076)
$\bar{R}^2$	0.426	0.381	0.453
Standardized effect on Δy_t			
Δs_{t-1}	-0.365	—	-0.396
r_t^{SP}	—	0.299	0.087

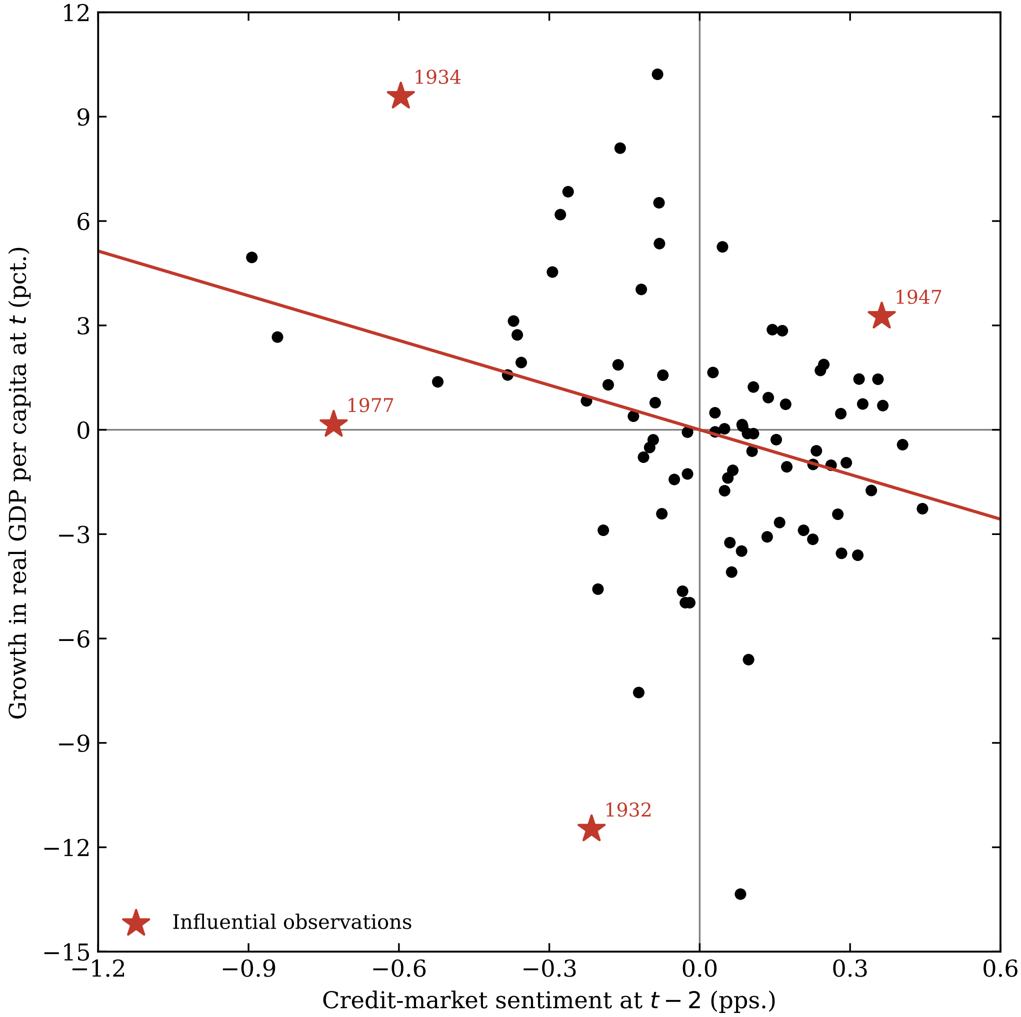

Figure 6: Figure II (replication): the credit-market sentiment measure.

Table II (replication).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.281***	—	-3.861***	-4.907***
	(1.253)		(1.301)	(1.678)
$\hat{r}_t^{SP}$	—	0.145*	0.053	—
		(0.082)	(0.074)	
Δy_{t-1}	0.597***	0.534***	0.590***	0.581***
	(0.114)	(0.099)	(0.112)	(0.096)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.099
				(0.313)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.616
				(0.414)
π_{t-1}	—	—	—	0.123
				(0.154)
R^2	0.377	0.337	0.380	0.394

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \mathrm{HYS}_{t-2}$	0.147***	—
	(0.042)	
s_{t-2}	-0.245***	—
	(0.047)	
$\ln[P/E10]_{t-2}$	—	-0.132***
		(0.040)
R^2	0.102	0.083

5 Extended Sample (1929–2025)

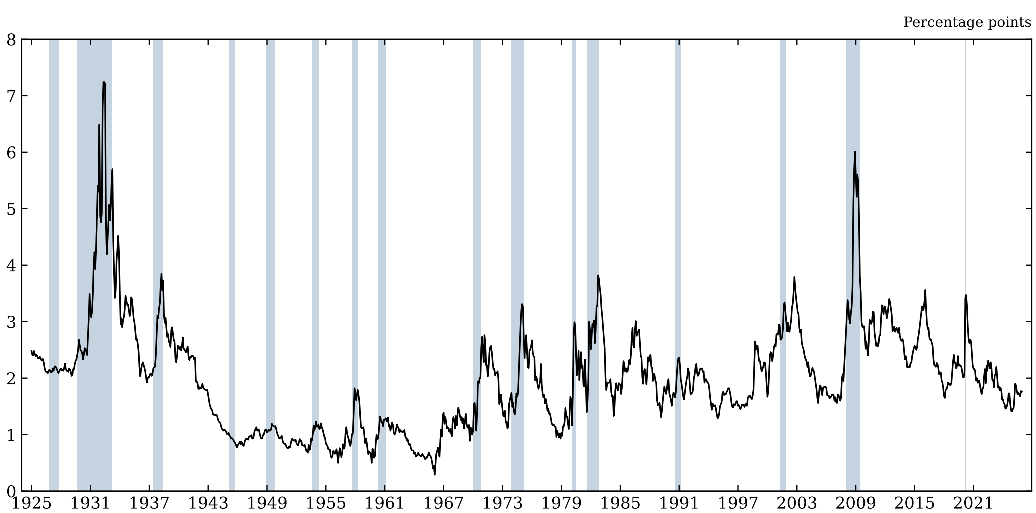

Figure 7: Figure I extended to the present.

Table I (extended).

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.871***	—	-2.033***
	(0.566)		(0.571)
r_t^{SP}	—	0.069**	0.017
		(0.032)	(0.034)
Δy_{t-1}	0.461***	0.465***	0.449***
	(0.108)	(0.136)	(0.116)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.170
			(0.265)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.776**
			(0.358)
π_{t-1}	—	—	0.089
			(0.078)
$\bar{R}^2$	0.395	0.352	0.415
Standardized effect on Δy_t			
Δs_{t-1}	-0.350	—	-0.380
r_t^{SP}	—	0.283	0.068

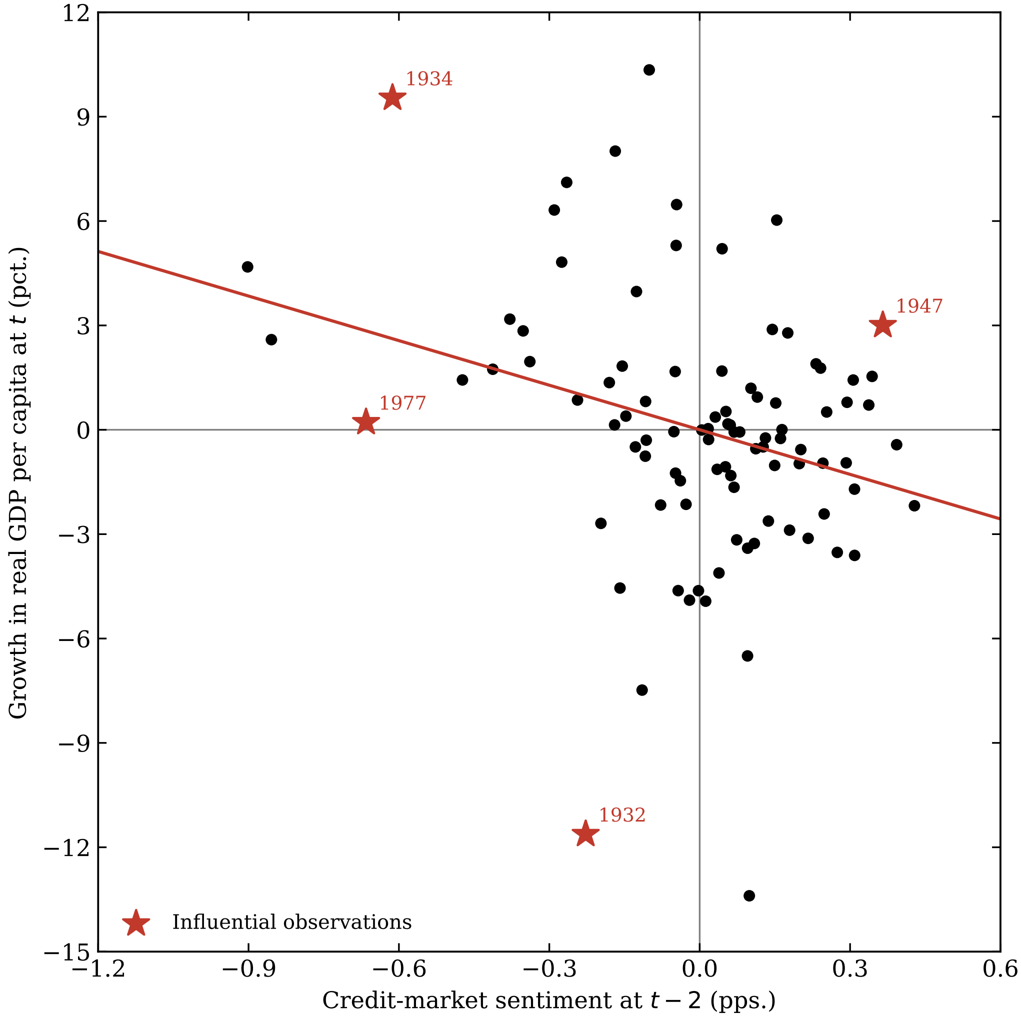

Figure 8: Figure II extended to the present.

Table II (extended).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.271***	—	-3.884***	-4.900***
	(1.287)		(1.307)	(1.679)
$\hat{r}_t^{SP}$	—	0.176*	0.068	—
		(0.101)	(0.089)	
Δy_{t-1}	0.577***	0.516***	0.570***	0.559***
	(0.118)	(0.103)	(0.115)	(0.099)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.117
				(0.286)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.603
				(0.402)
π_{t-1}	—	—	—	0.121
				(0.154)
R^2	0.353	0.313	0.356	0.368

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \text{HYS}_{t-2}$	0.130***	—
	(0.039)	
s_{t-2}	-0.244***	—
	(0.046)	
$\ln[P/E10]_{t-2}$	—	-0.091***
		(0.035)
R^2	0.096	0.047

6 Extension: Aaa Credit Spread

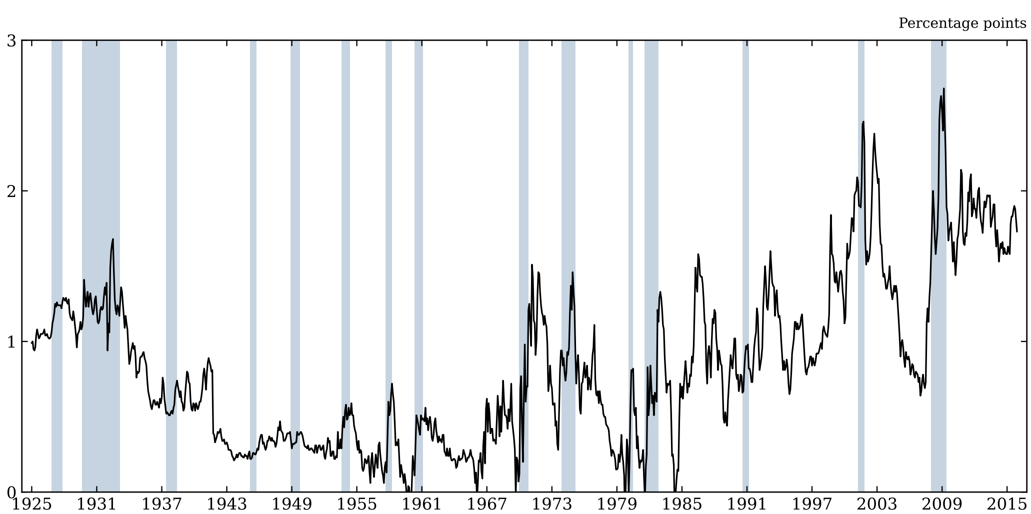

Figure 9: Figure I, Aaa spread (replication sample).

Table I, Aaa (replication).

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.958***	—	-3.192***
	(0.734)		(1.168)
r_t^{SP}	—	0.075**	0.056
		(0.033)	(0.035)
Δy_{t-1}	0.517***	0.478***	0.477***
	(0.109)	(0.136)	(0.121)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.326
			(0.314)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.917*
			(0.504)
π_{t-1}	—	—	0.102
			(0.108)
$\bar{R}^2$	0.321	0.381	0.396
Standardized effect on Δy_t			
Δs_{t-1}	-0.165	—	-0.269
r_t^{SP}	—	0.299	0.225

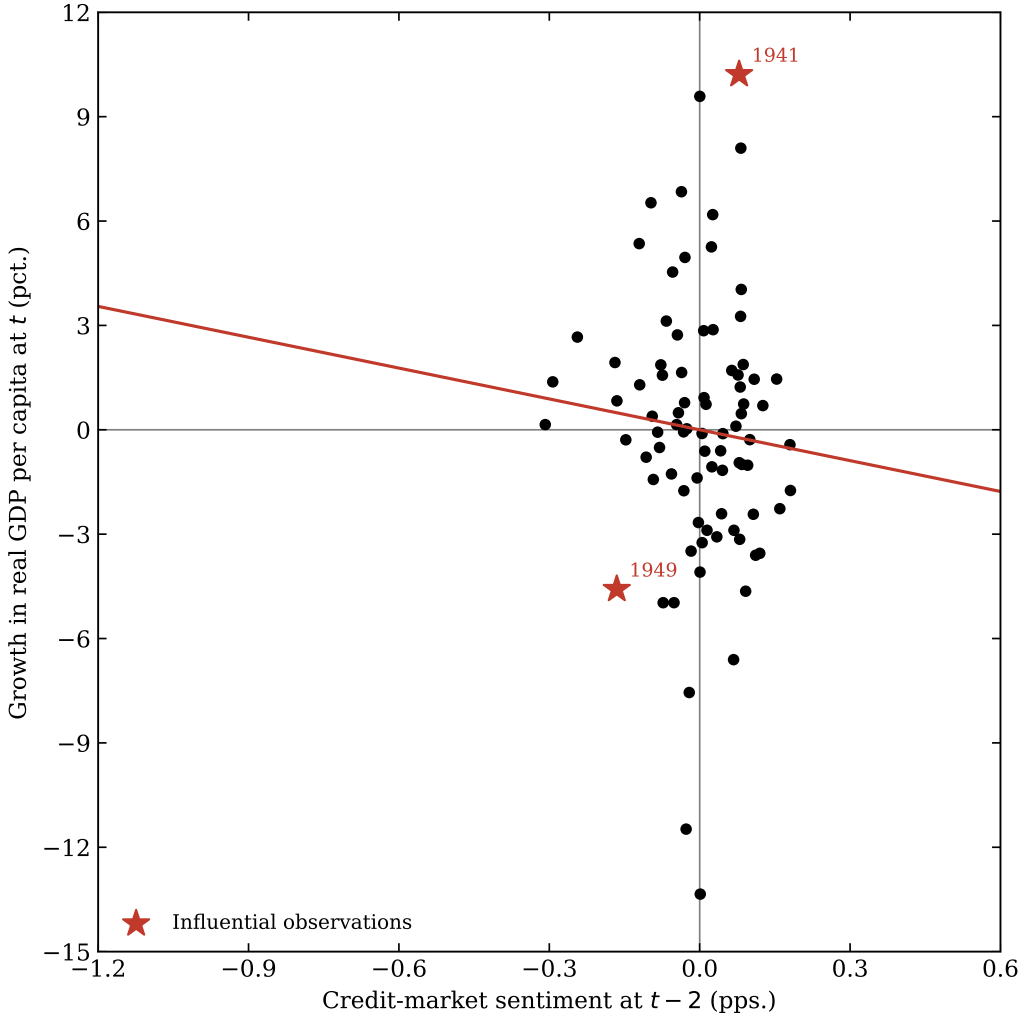

Figure 10: Figure II, Aaa spread (replication sample).

Table II, Aaa (replication).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.956	—	-2.362	-3.663
	(2.488)		(2.865)	(2.806)
$\hat{r}_t^{SP}$	—	0.145*	0.131	—
		(0.082)	(0.085)	
Δy_{t-1}	0.555***	0.534***	0.549***	0.559***
	(0.108)	(0.099)	(0.100)	(0.105)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.063
				(0.308)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.373
				(0.394)
π_{t-1}	—	—	—	0.043
				(0.132)
R^2	0.321	0.337	0.342	0.328

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \mathrm{HYS}_{t-2}$	0.078***	—
	(0.029)	
s_{t-2}	-0.167***	—
	(0.058)	
$\ln[P/E10]_{t-2}$	—	-0.132***
		(0.040)
R^2	0.064	0.083

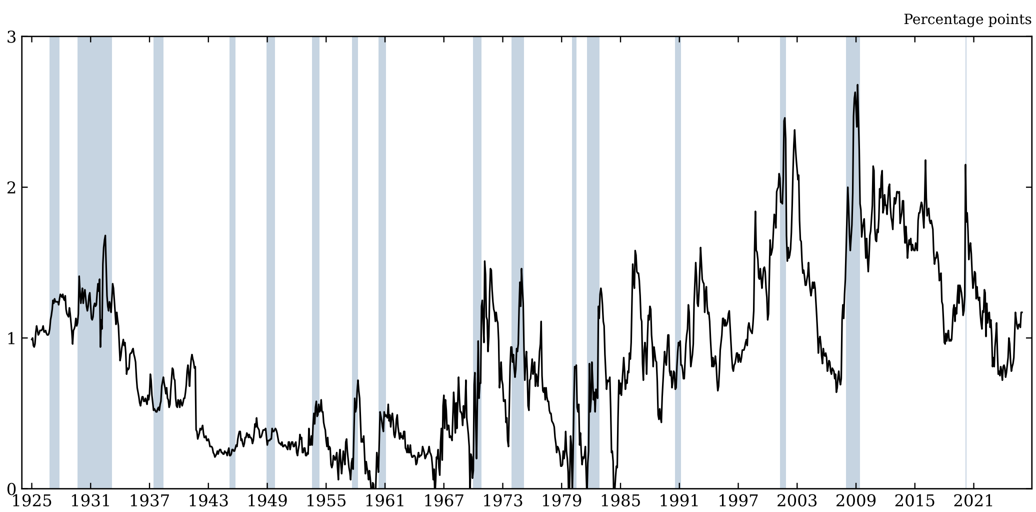

Figure 11: Figure I, Aaa spread (extended sample).

Table I, Aaa (extended).

	Dependent variable: Δy_t		
Regressors	(1)	(2)	(3)
Δs_{t-1}	-1.812**	—	-2.926**
	(0.713)		(1.142)
r_t^{SP}	—	0.069**	0.050
		(0.032)	(0.034)
Δy_{t-1}	0.500***	0.465***	0.463***
	(0.112)	(0.136)	(0.124)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.246
			(0.285)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.829*
			(0.478)
π_{t-1}	—	—	0.092
			(0.108)
$\bar{R}^2$	0.298	0.352	0.360
Standardized effect on Δy_t			
Δs_{t-1}	-0.154	—	-0.248
r_t^{SP}	—	0.283	0.206

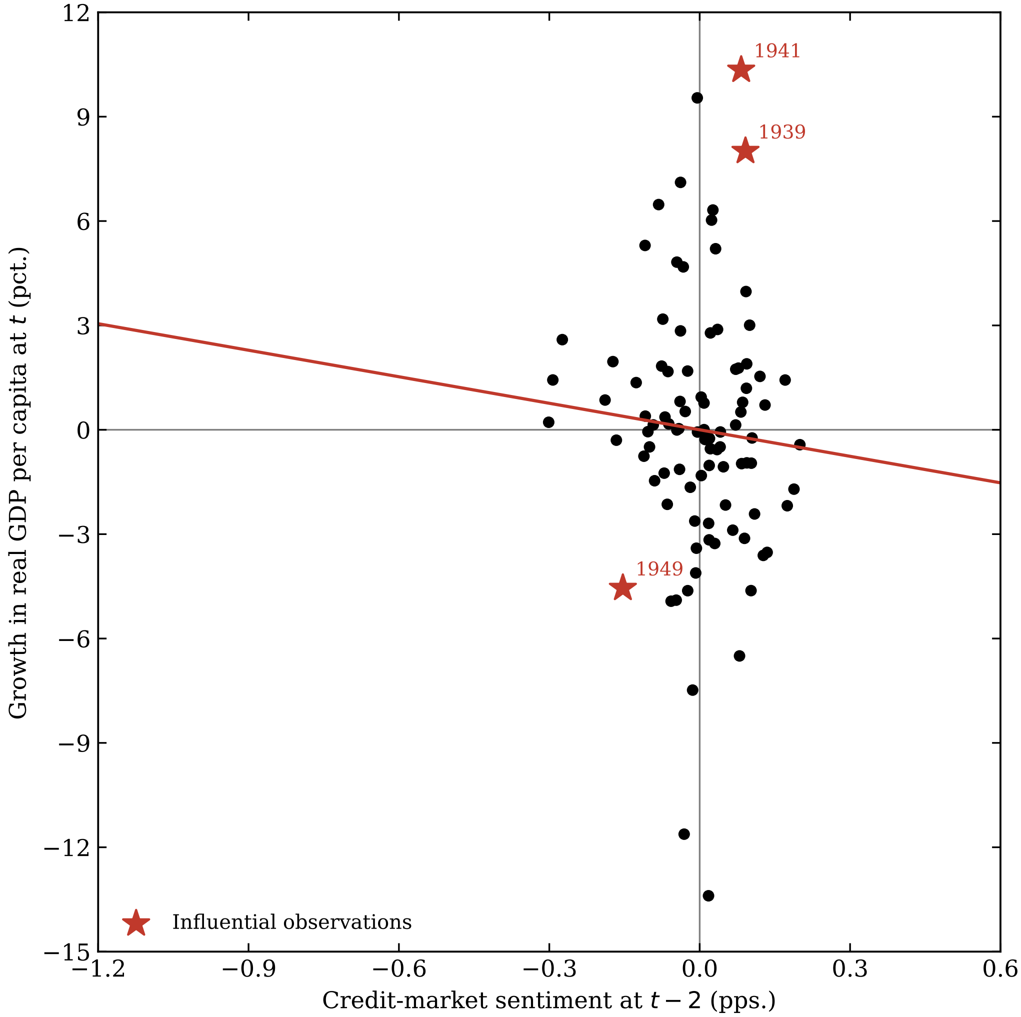

Figure 12: Figure II, Aaa spread (extended sample).

Table II, Aaa (extended).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.541	—	-2.489	-3.195
	(2.342)		(2.532)	(2.652)
$\hat{r}_t^{SP}$	—	0.176*	0.162	—
		(0.101)	(0.104)	
Δy_{t-1}	0.536***	0.516***	0.532***	0.540***
	(0.111)	(0.103)	(0.103)	(0.107)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.047
				(0.281)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.372
				(0.382)
π_{t-1}	—	—	—	0.041
				(0.132)
R^2	0.298	0.313	0.318	0.306

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \mathrm{HYS}_{t-2}$	0.077***	—
	(0.028)	
s_{t-2}	-0.185***	—
	(0.058)	
$\ln[P/E10]_{t-2}$	—	-0.091***
		(0.035)
R^2	0.070	0.047

7 Case Study: The Sentiment Signal Out of Sample, 2020–2022

The post-2008 reconstruction of the high-yield issuance share from Mergent FISD makes it possible to apply the Table II first stage, estimated on 1929–2015 and held fixed, to the COVID cycle. The figure below compares the predicted change in the Baa–Treasury spread—using the full two-ingredient sentiment signal and, separately, the spread-reversion leg alone—against the realized change. The full walkthrough, including the caveats that bound how much weight a three-year window can carry, is in `04_case_study.ipynb`.

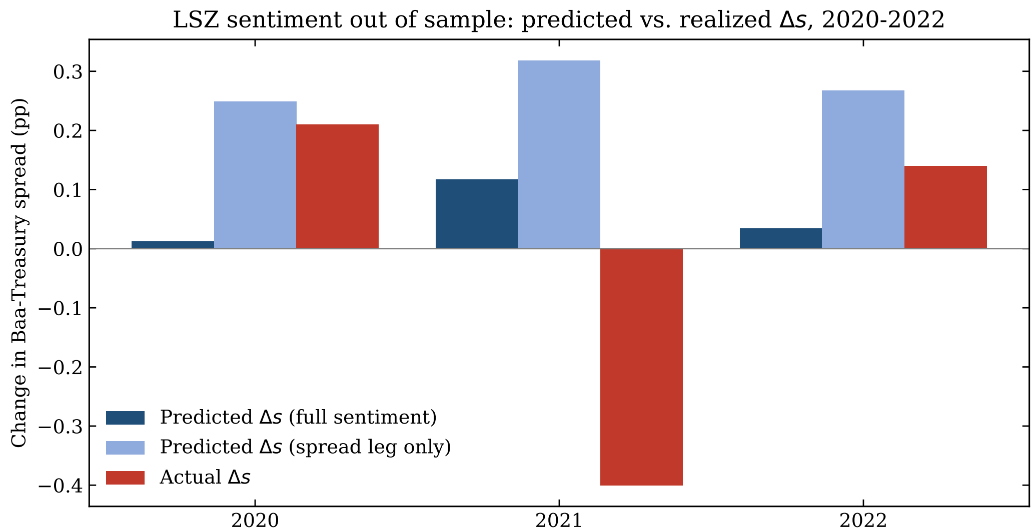

Figure 13: Out-of-sample application of the López-Salido, Stein & Zakrajšek sentiment first stage to 2020–2022: predicted versus realized change in the credit spread.

8 Successes and Challenges

Where it worked. The long-run credit spread (Figure I) reproduces cleanly from FRED and shows the expected countercyclical spikes into every recession. The central result held in Table II: changes in the fitted credit spread forecast subsequent output growth (main specification $R^2 \approx 0.38$), and the auxiliary regression confirms the lagged log high-yield share and lagged spread jointly predict spread changes. Extending the sample to the present preserved the qualitative result.

Where it was hard. The main difficulty was the Greenwood–Hanson high-yield issuance share, which is not a clean FRED pull; sourcing it reliably was the chief data bottleneck, resolved by pulling the published workbook directly from its URL and splicing it with a Mergent FISD reconstruction via WRDS. Assembling the FRED series was a second source of friction: several series had to be stitched together to form a continuous long history, and the paper's appendix does not fully pin down which underlying series were used, so some choices were our own documented judgement calls. Matching the paper's inference also required Newey–West standard errors with a suitable lag, and differences in sample period and data vintage shift point estimates relative to the published tables—expected, and explained by the updated data vintages and series choices noted above.